Light

GCSE Computer Science — CPU Architecture (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main function of the Arithmetic Logic Unit (ALU) in a CPU?

- ○ **A.** To store instructions waiting to be executed
- ○ **B.** To perform calculations and logical comparisons
- ○ **C.** To control the timing of CPU operations
- ○ **D.** To decode instructions fetched from memory

Q2 (1 mark)

Which CPU component is responsible for coordinating all activities within the processor?

- ○ **A.** The control unit

- ○ **B.** The arithmetic logic unit
- ○ **C.** The system clock
- ○ **D.** The data bus

Q3 (1 mark)

What is the purpose of the clock in a CPU?

- ○ **A.** To store the current time and date for the operating system
- ○ **B.** To transfer data between the CPU and main memory
- ○ **C.** To synchronise the timing of all CPU operations
- ○ **D.** To calculate how long each instruction takes to execute

Q4 (1 mark)

Which statement best describes a register in a CPU?

- ○ **A.** A large storage area for programs and data
- ○ **B.** A component that performs arithmetic calculations
- ○ **C.** A device that connects the CPU to external peripherals
- ○ **D.** A small, fast storage location within the processor

Q5 (1 mark)

What is a bus in a computer system?

- ○ **A.** A collection of wires that transmit data and signals between components
- ○ **B.** A type of processor used for graphics rendering
- ○ **C.** Software that manages communication between programs
- ○ **D.** A cooling system that prevents the CPU from overheating

Q6 (1 mark)

Clock speed is typically measured in which unit?

- ○ **A.** Megabytes (MB)
- ○ **B.** Hertz (Hz)
- ○ **C.** Bits per second (bps)
- ○ **D.** Watts (W)

Q7 (1 mark)

What is the primary purpose of cache memory?

- ○ **A.** To permanently store the operating system
- ○ **B.** To provide long-term storage for user files
- ○ **C.** To store frequently accessed data for faster retrieval by the CPU
- ○ **D.** To back up important data in case of power failure

Q8 (1 mark)

What does it mean when a CPU is described as 'quad-core'?

- ○ **A.** It has four levels of cache memory
- ○ **B.** It can execute four instructions per clock cycle
- ○ **C.** It operates at four times the normal clock speed
- ○ **D.** It contains four independent processing units

Q9 (1 mark)

What is the function of the address bus?

- ○ **A.** To carry memory addresses from the CPU to identify locations in memory
- ○ **B.** To transfer the actual data between the CPU and memory
- ○ **C.** To send control signals that manage read and write operations
- ○ **D.** To provide power to memory modules

Q10 (1 mark)

Which bus is responsible for transferring the actual data between the CPU and main memory?

- ○ **A.** The address bus
- ○ **B.** The control bus
- ○ **C.** The data bus

- ○ **D.** The system bus

Q11 (1 mark)

What type of signals does the control bus carry?

- ○ **A.** Memory addresses to locate data
- ○ **B.** Signals such as read, write, and interrupt commands
- ○ **C.** Binary data being processed by the ALU
- ○ **D.** Clock pulses to synchronise components

Q12 (1 mark)

Which characteristic best describes RAM (Random Access Memory)?

- ○ **A.** It permanently stores the computer's startup instructions
- ○ **B.** It can only be read, not written to, during normal operation
- ○ **C.** It retains data even when the computer is switched off
- ○ **D.** It is volatile and loses its contents when power is removed

Q13 (2 marks)

A computer has a CPU with a clock speed of 3.2 GHz. A newer CPU model has a clock speed of 4.0 GHz. Which statement best explains how this affects performance?

- ○ **A.** The faster clock speed allows more instructions to be processed per second, improving overall performance
- ○ **B.** The faster clock speed increases the amount of RAM available to programs
- ○ **C.** The faster clock speed reduces the number of cores needed in the processor
- ○ **D.** The faster clock speed increases the size of the cache memory

Q14 (2 marks)

Why is cache memory faster than RAM for the CPU to access?

- ○ **A.** Cache uses a different type of electricity that travels faster
- ○ **B.** Cache is located closer to the CPU and uses faster memory technology
- ○ **C.** Cache can store more data than RAM
- ○ **D.** Cache is connected directly to the power supply

Q15 (2 marks)

A video editing application is designed to use multiple processor cores. Why would a quad-core processor perform better than a dual-core processor for this application?

- ○ **A.** Quad-core processors always have faster clock speeds
- ○ **B.** Quad-core processors have more cache memory by default
- ○ **C.** Four cores can process different parts of the video simultaneously, reducing total processing time

○ **D.** Quad-core processors use less power, allowing faster processing

Q16 (2 marks)

A programmer wants to understand why accessing data from a register is faster than accessing data from RAM. Which explanation is correct?

○ **A.** Registers store data in a compressed format that is quicker to decompress

○ **B.** Registers are larger than RAM, so they can hold more data at once

○ **C.** RAM requires the data to pass through the cache before reaching the CPU

○ **D.** Registers are located inside the CPU, eliminating the need to use buses

Q17 (2 marks)

A computer has a 32-bit address bus. What is the maximum amount of memory this computer can directly address?

○ **A.** 32 megabytes (MB)

○ **B.** 4 gigabytes (GB)

○ **C.** 32 gigabytes (GB)

○ **D.** 4 terabytes (TB)

Q18 (2 marks)

A program needs to check if a user's age (stored as 17) is greater than or equal to 18. Which CPU component performs this comparison, and what type of operation is it?

- ○ **A.** The ALU performs a logical comparison operation
- ○ **B.** The control unit performs an arithmetic operation
- ○ **C.** The cache performs a data retrieval operation
- ○ **D.** The register performs a storage operation

Q19 (2 marks)

During the execution of a program, the control unit decodes an instruction that requires adding two numbers. What happens next?

- ○ **A.** The control unit performs the addition directly
- ○ **B.** The instruction is sent back to main memory for processing
- ○ **C.** The cache stores the result before the addition is performed
- ○ **D.** The control unit sends signals to the ALU to perform the addition

Q20 (2 marks)

Computer A has a single-core 4.0 GHz processor. Computer B has a dual-core 2.5 GHz processor. For a program that can only use one core, which computer will likely complete the task faster?

- ○ **A.** Computer B, because it has two cores available
- ○ **B.** Both computers will perform equally because the total GHz is similar

- ○ **C.** Computer A, because single-threaded programs benefit from higher clock speed
- ○ **D.** Computer B, because dual-core processors are always faster

Q21 (2 marks)

A company is choosing between two processors with identical clock speeds and core counts. Processor X has 8 MB of cache while Processor Y has 16 MB of cache. Why might Processor Y offer better performance?

- ○ **A.** Larger cache can store more frequently used data, reducing the need to access slower RAM
- ○ **B.** Larger cache increases the clock speed of the processor
- ○ **C.** Larger cache allows the processor to have more cores
- ○ **D.** Larger cache reduces the voltage required by the processor

Q22 (2 marks)

A user is running a web browser, a music streaming application, and an antivirus scan simultaneously. Why would a multi-core processor be beneficial in this situation?

- ○ **A.** Each application requires a dedicated core to function at all
- ○ **B.** Each core can handle a different application, allowing true parallel processing
- ○ **C.** Multi-core processors have faster internet connectivity
- ○ **D.** Multi-core processors have more RAM available for applications

Q23 (2 marks)

When the CPU reads data from a specific memory location, which sequence correctly describes how the three main buses work together?

- ○ **A.** Data bus sends the address, control bus fetches the data, address bus signals completion
- ○ **B.** Control bus sends the address, data bus receives the data, address bus signals read operation
- ○ **C.** Address bus sends the memory address, control bus signals a read operation, data bus carries the retrieved data
- ○ **D.** Address bus carries the data, data bus specifies the address, control bus stores the result

Q24 (3 marks)

A school is purchasing new computers for its computer science department. They must choose between two models:

- **Model P:** 3.0 GHz quad-core processor, 8 MB cache
- **Model Q:** 4.0 GHz dual-core processor, 4 MB cache

The computers will primarily run multi-threaded software development tools. Which model is likely to offer better overall performance for this use case, and why?

- ○ **A.** Model Q, because the higher clock speed will make all operations faster regardless of core count
- ○ **B.** Model P, because the software can only use one core, making cache size the deciding factor
- ○ **C.** Model Q, because dual-core processors are more reliable than quad-core processors
- ○ **D.** Model P, because multi-threaded software benefits from more cores and larger cache compensates for lower clock speed

Q25 (3 marks)

A CPU has a 16-bit data bus and operates at a clock speed where it can complete one memory read operation per clock cycle. The CPU needs to load a 64-bit value from memory into a register.
What is the minimum number of memory read operations required, and why?

- ○ **A.** Four read operations, because only 16 bits can be transferred per read cycle
- ○ **B.** One read operation, because the register can request all 64 bits at once
- ○ **C.** Two read operations, because the bus can double its width for large transfers
- ○ **D.** Eight read operations, because each bit requires a separate read cycle

Q26 (3 marks)

A user complains their computer is slow when running multiple applications. System monitoring shows:

- CPU usage: 95% average
- RAM usage: 45% of 16 GB
- Storage read speed: 150 MB/s (SSD rated at 500 MB/s)

Which upgrade would most effectively improve the computer's performance for this usage pattern?

- ○ **A.** Installing a faster SSD to increase storage read speeds
- ○ **B.** Upgrading to a CPU with more cores or higher clock speed
- ○ **C.** Adding more RAM to increase available memory
- ○ **D.** Replacing the motherboard to improve bus speeds

Q27 (3 marks)

A modern CPU has three levels of cache memory:

- L1 cache: 64 KB, access time 1 nanosecond
- L2 cache: 512 KB, access time 4 nanoseconds
- L3 cache: 8 MB, access time 12 nanoseconds

RAM access time is 100 nanoseconds. Why is this hierarchical cache structure more effective than having a single large 8.5 MB cache?

- ○ **A.** Multiple cache levels use less power than a single large cache
- ○ **B.** A single large cache would be too hot and require additional cooling
- ○ **C.** Frequently accessed data in L1 cache can be retrieved faster than if all cache operated at L3 speeds
- ○ **D.** Multiple cache levels allow different applications to use different cache levels simultaneously

Q28 (3 marks)

A laptop manufacturer must choose between two processor designs for a new model:

- **Design A:** 6 cores at 2.4 GHz, 15W power consumption
- **Design B:** 4 cores at 3.6 GHz, 25W power consumption

For a laptop prioritising battery life while still offering good performance for office applications (which are typically single-threaded), which design is the better choice?

- ○ **A.** Design A, because more cores always provide better performance
- ○ **B.** Design B, because office applications run faster on the higher clock speed core, but battery life will suffer
- ○ **C.** Design A, because lower power consumption extends battery life, though single-threaded performance will be slightly lower

○ **D.** Design B, because the extra power consumption is offset by faster task completion

Q29 (3 marks)

A computer system is analysed and the following timings are recorded for the CPU's fetch-decode-execute cycle:

- Fetch instruction from memory: 20 nanoseconds
- Decode instruction: 2 nanoseconds
- Execute instruction: 3 nanoseconds

Which component represents the bottleneck in this system, and what upgrade would be most effective?

○ **A.** Memory access is the bottleneck; increasing cache size or using faster RAM would be most effective

○ **B.** The ALU is the bottleneck; upgrading to a CPU with faster execution capability would be most effective

○ **C.** The control unit is the bottleneck; upgrading to a CPU with better instruction decoding would be most effective

○ **D.** The clock speed is the bottleneck; increasing the CPU frequency would be most effective

Q30 (3 marks)

A gaming PC manufacturer is designing a high-performance system. They can choose one of the following configurations:

- **Config X:** 8-core CPU at 5.0 GHz, 32 MB L3 cache, DDR5 RAM
- **Config Y:** 16-core CPU at 3.5 GHz, 64 MB L3 cache, DDR5 RAM

Modern games typically use 4—6 cores effectively, with some using up to 8 cores. Single-threaded performance significantly impacts game responsiveness (frame timing). Which configuration is likely better for gaming, and why?

○ **A.** Config Y, because it has double the cores and cache, ensuring future-proofing for upcoming games

○ **B.** Config X, because higher clock speed improves single-threaded performance, and most games cannot effectively use more than 8 cores

○ **C.** Config Y, because the larger cache compensates for any clock speed disadvantage

○ **D.** Both configurations would perform identically because they have the same RAM type

Total: **55** marks • Time: **30 min**

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